Sharp One-Dimensional Lieb–Thirring Constants
Quadratic Cutoff-Gram Errors and Quantitative Dirichlet Exhaustion

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Abstract

The source's scalar all-exponent inequality passes from bounded intervals to the line through finite cutoff eigenfunction spaces. We retain the exact common-cutoff localization identity instead of treating its Gram and form errors independently. Physical normalization then leaves a double commutator with a sharp abstract matrix bound depending only on the spectral width. Its Gram contribution is quadratic near the identity and vanishes on an equal-energy space. Angular cutoffs tie the remaining gradient cost to the actual lost mass. At exponent one an exact trace identity removes the commutator cost altogether. We give finite-list moment consequences, a form-dual residual extension, an explicit one-state Dirichlet enclosure, and counterexamples to unconditional spectral completion. The source's matrix-field, rank-one and connecting-path machinery remains an imported route to its sharp global constant. Classical localization and commutator estimates are credited; the improvement is a quantitative refinement of this specific exhaustion step, not another proof of the global conjecture. Contribution assessment: C3/5, sharp spectral-width cutoff-Gram control and trace-exact moment transfer strengthen the source's finite exhaustion estimates without certifying the global action inequality or unseen spectral tails.

1 The scalar problem and its domains

Let

$$\tfrac{1}{2} < \gamma < \tfrac{3}{2}, \qquad \sigma = \gamma - \tfrac{1}{2} \in (0,1), \quad p = 1+\sigma, \quad q = 1+\sigma^{-1}.$$

The source [1] claims, for every $0 \le W \in L^p(\mathbb{R})$,

$$\operatorname{Tr}(-\partial_x^2 - W)_-^\gamma \le L_\gamma \int_{\mathbb{R}} W^p, \qquad L_\gamma = \frac{\sigma^\sigma}{(1+\sigma)^{1+\sigma}\mathrm{B}(1/2, 1+\sigma)}. \tag{1}$$

Every negative eigenvalue, with multiplicity, is included. Its sharpness example is $W = (r+1)\operatorname{sech}^2(rx)$, $r = 1/\sigma$. The manuscript exhibits attainment, not a classification of all optimizers. Its matrices encode finite scalar families; they do not change the physical Hilbert space into a matrix-valued potential problem.

We use inner products conjugate linear in the first slot. More generally let V be real measurable, $V_- \in L^p(\mathbb{R})$, and assume

$$\mathcal{D}_V = \{u \in H^1(\mathbb{R}) : \int_{\mathbb{R}} V_+ |u|^2 < \infty\}$$

is dense in L^2. The local condition $V_+ \in L^1_{\mathrm{loc}}$ is sufficient. The form and associated operator are

$$h_V[u,v] = \int \overline{u'}v' + \int V\overline{u}v, \qquad D(h_V) = \mathcal{D}_V, \tag{2}$$

$$D(H_V) = \{u \in \mathcal{D}_V : \exists f \in L^2,\ h_V[v,u] = \langle v, f\rangle\ \forall v \in \mathcal{D}_V\}. \tag{3}$$

The negative-part integral is finite on this domain. Indeed, Sobolev interpolation and Young's inequality imply, for every $\varepsilon > 0$,

$$\int V_-|u|^2 \le 2^{1/p}\,\|V_-\|_p \left\|u'\right\|_2^{1/p} \|u\|_2^{2-1/p} \le \varepsilon \left\|u'\right\|_2^2 + C_\varepsilon \|u\|_2^2 .$$

The positive weighted form is closed; this infinitesimal form bound gives a closed semibounded form and a self-adjoint H_V. Multiplication by $\sqrt{V_-}$ is compact from H^1 to L^2, by bounded compactly supported approximation in L^p and local Rellich compactness. Thus negative spectral projections bounded away from zero have finite rank. On a bounded interval I, the Dirichlet form domain is

$$\mathcal{D}_{V,I} = H^1_0(I) \cap L^2(I, V_+\,dx);$$

zero extension embeds it into $\mathcal{D}_V$. Its form embedding in $L^2(I)$ is compact, so the Dirichlet operator has compact resolvent. Density follows by multiplying approximations from $\mathcal{D}_V$ by cutoffs supported inside I.

There is no finite-boundary condition on the full line and no gauge quotient. Homogeneous Dirichlet traces are imposed only after cutoff. No H^2 regularity is assumed for rough V. Everything below uses forms and bounded H^1 multipliers, rather than unjustified pointwise differential equations.

2 The exact common-cutoff identity

Fix $N < \infty$ orthonormal eigenfunctions $\phi_1, \dots, \phi_N \in D(H_V)$, with real eigenvalues $\lambda_1 \le \cdots \le \lambda_N$; in the spectral application these are the first N negative eigenvalues. Write $\Lambda = \mathrm{diag}(\lambda_i)$. Let $\chi \in W^{1,\infty}(\mathbb{R})$ be real, $0 \le \chi \le 1$, and zero outside a bounded interval I. Define

$$G_{ij} = \int \chi^2 \overline{\phi_i}\phi_j, \quad Q_{ij} = h_V[\chi\phi_i, \chi\phi_j], \quad E_{ij} = \int |\chi'|^2 \overline{\phi_i}\phi_j. \tag{4}$$

Both G and E are physical integral matrices; no normalization by N is used. We always have $0 \le G \le I_N$ and $E \ge 0$.

Lemma 2.1 (Polarized localization)**.** *For the preceding data,*

$$Q = \tfrac{1}{2}(\Lambda G + G\Lambda) + E. \tag{5}$$

Proof. The product rule for H^1 functions and a Lipschitz multiplier gives

$$h_V[\chi u, \chi v] = \tfrac{1}{2}\{h_V[\chi^2 u, v] + h_V[u, \chi^2 v]\} + \int |\chi'|^2 \overline{u}v.$$

The mixed derivative terms cancel directly. Multiplication by χ and χ^2 preserves $\mathcal{D}_V$, including its positive weighted part, so every pairing is legitimate. The weak eigenfunction equation gives $h_V[\chi^2\phi_i, \phi_j] = \lambda_j G_{ij}$; Hermitian symmetry gives the other term $\lambda_i G_{ij}$. This proves the matrix identity, without differentiating V or assuming that $\chi\phi_i \in D(H_V)$. □

This is the classical IMS localization calculation, here kept as a full matrix identity. A recent primary proof of the quadratic-form version is [3, Appendix A, Lemma A.3]. The matrix consequence does *not* say $Q = \Lambda G + E$: in general G and Λ do not commute.

3 A sharp spectral-width normalization bound

The source already uses

$$\eta_{\text{source}} = \frac{e + \|\Lambda\|_{\text{op}}\, g}{1-g}, \qquad e = \|Q-\Lambda\|_{\text{op}}, \quad g = \|I-G\|_{\text{op}} < 1. \tag{6}$$

That estimate is valid for independently controlled form and Gram errors. It is not new here. Common cutoff gives additional structure. Assume $G > 0$, set $S = G^{-1/2}$, and let $\widehat{Q} = SQS$ be the form in the orthonormalized physical basis.

Theorem 3.1 (Quadratic Gram cost with sharp abstract coefficient)**.** *Let $mI \le G \le MI$, $0 < m \le M$, and let $\Lambda = \Lambda^*$ have spectral width $\Delta = \lambda_{\max}(\Lambda) - \lambda_{\min}(\Lambda)$. If $Q = (\Lambda G + G\Lambda)/2 + E$, $E = E^*$, then*

$$\widehat{Q} - \Lambda = SES + \tfrac{1}{2}[S,[\Lambda, S^{-1}]], \tag{7}$$

$$\left\|\widehat{Q} - \Lambda\right\|_{\text{op}} \le \|SES\|_{\text{op}} + \frac{\Delta(\sqrt{M}-\sqrt{m})^2}{4\sqrt{mM}}. \tag{8}$$

The second term has best possible coefficient in this abstract matrix class. If $0 < G \le I$ and $g = \|I-G\|_{\text{op}} < 1$, it is sufficient to use

$$\eta = \frac{\|E\|_{\text{op}}}{1-g} + \frac{\Delta(1-\sqrt{1-g})^2}{4\sqrt{1-g}}. \tag{9}$$

The Gram term is $\Delta g^2/16 + O(\Delta g^3)$ as $g \to 0$.

Proof. Since $GS = S^{-1}$,

$$SQS = \tfrac{1}{2}(S\Lambda S^{-1} + S^{-1}\Lambda S) + SES.$$

Expanding the double commutator proves (7). Scalar shifts do not affect a commutator. Center Λ at its spectral midpoint, S at $(m^{-1/2} + M^{-1/2})/2$, and S^{-1} at $(\sqrt{m} + \sqrt{M})/2$. Their centered norms are bounded respectively by

$$\Delta/2, \qquad (m^{-1/2} - M^{-1/2})/2, \qquad (\sqrt{M} - \sqrt{m})/2.$$

Using $\|[A,B]\| \le 2\,\|A\|\,\|B\|$ twice and the factor $1/2$ outside the double commutator gives exactly the second term of (8). The first term is at most $\|E\|/m$. For a cutoff, take $m = 1-g$, $M = 1$. Finally, $(1-\sqrt{1-g})^2 = g^2/(1+\sqrt{1-g})^2$, which gives the expansion. □

Example 3.2 (Sharpness and shift invariance)**.** Let $0 < m < M$, $\Delta > 0$, and

$$G = \operatorname{diag}(m, M), \qquad \Lambda = \ell I + \frac{\Delta}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad E = 0.$$

Then $\widehat{Q} - \Lambda$ is the off-diagonal matrix with both off-diagonal entries $\Delta(\sqrt{M}-\sqrt{m})^2/(4\sqrt{mM})$. It attains (8), and its positive eigenvector also gives the top eigenvalue displacement by that amount. A constant unitary diagonalizes Λ without changing this assertion. Choosing $\ell < -\Delta/2$ puts both original eigenvalues below zero. This is a sharpness example for the abstract matrix estimate, not a claim that these exact matrices with $E = 0$ are realized by compactly cut off Schrödinger eigenfunctions.

Replacing Λ by $\Lambda + cI$ shifts $\widehat{Q}$ by the same cI; the error bound is unchanged. It vanishes in its Gram part when $\Delta = 0$, including $N = 1$. In contrast (6) pays an absolute-energy term even for one state. The classical commutator inequalities are not new; the useful increment is the exact cutoff structure, spectral centering and sharp coefficient at this source step.

4 Angular cutoffs and quantitative exhaustion

Let $\chi = \cos\theta$, where $\theta \in W^{1,\infty}(\mathbb{R})$, $0 \le \theta \le \pi/2$, and $\theta = \pi/2$ outside I. If $\|\theta'\|_\infty \le L$, then

$$E \le L^2(I - G), \qquad SES \le L^2(G^{-1} - I). \tag{10}$$

Indeed $|\chi'|^2 = |\theta'|^2 \sin^2\theta \le L^2(1 - \chi^2)$; integration against the square of every linear combination of the eigenfunctions proves the matrix order. This is stronger than a bound using the full L^2 mass.

Corollary 4.1 (A finite-list spectral enclosure)**.** *Assume the functions in Section 2 are the first N eigenfunctions below zero and $g < 1$. Put*

$$\eta_\chi = \frac{L^2 g}{1 - g} + \frac{\Delta(1 - \sqrt{1 - g})^2}{4\sqrt{1 - g}}. \tag{11}$$

Then for every $j \le N$,

$$\lambda_j \le \lambda_j(I) \le \lambda_j + \eta_\chi. \tag{12}$$

In particular, all these Dirichlet levels are negative if $\eta_\chi < -\lambda_N$.

Proof. Zero extension of Dirichlet form vectors gives the lower inequality by min–max. The normalized cutoff space is N-dimensional. Its ordered Ritz values μ_j obey $\lambda_j(I) \le \mu_j$. The finite Hermitian min–max principle and (8) imply $|\mu_j - \lambda_j| \le \eta_\chi$. Combining these facts proves the assertion. Neither a full-line eigenfunction nor a cutoff is assumed to have pointwise second derivatives. □

For example, take $\theta = 0$ on $[-R, R]$, increase linearly to $\pi/2$ across each adjacent interval of length $w > 0$, and set it equal to $\pi/2$ outside. This continuous piecewise linear function is in $W^{1,\infty}$, with $L = \pi/(2w)$. Its cosine has zero endpoint trace on $I = (-R - w, R + w)$. The corners require no smoothing for the form argument. Smooth versions with a certified derivative cap work too.

A sufficient observable tail bound is

$$g \le \operatorname{tr}(I - G) = \int (1 - \chi^2) \sum_{j=1}^{N} |\phi_j|^2 \le \int_{|x|>R} \sum_{j=1}^{N} |\phi_j|^2 =: t_N(R). \tag{13}$$

If $t_N(R) < 1$, replacing g by $t_N(R)$ in (11) is safe: both scalar factors there increase in g. For fixed N, $t_N(R) \to 0$, and $w = R$ gives a quantitative realization of the source exhaustion limit. There is no claim of a rate uniform in N or in all L^p potentials. The actual tail estimate remains an input.

5 Trace cancellation and finite spectral moments

The error in (7) is not positive semidefinite in general. Its commutator part, however, has trace zero. Define the actual gradient cost

$$\tau_\chi = \operatorname{tr}(G^{-1}E) \ge 0. \tag{14}$$

Theorem 5.1 (Trace-exact transfer for the first eigenvalue list)**.** *For the exact first N eigenfunctions and common cutoff above, let $\mu_1 \le \cdots \le \mu_N$ be the Ritz values of $\widehat{Q}$. Then*

$$\mu_j \ge \lambda_j, \qquad \sum_{j=1}^{N} (\mu_j - \lambda_j) = \tau_\chi. \tag{15}$$

Consequently

$$0 \le \lambda_j(I) - \lambda_j \le \min\{\eta_\chi, \tau_\chi\} \quad (j \le N). \tag{16}$$

At exponent one, without needing $\eta_\chi < -\lambda_N$,

$$\operatorname{Tr}(H_{V,I})_- \geq \left(\sum_{j=1}^{N} (-\lambda_j) - \tau_\chi \right)_+ . \tag{17}$$

For an angular cutoff,

$$\tau_\chi \leq L^2 \operatorname{tr}(G^{-1} - I) \leq \frac{L^2 \operatorname{tr}(I - G)}{1 - g}. \tag{18}$$

Proof. The cutoff space is a conforming subspace of the full-line form domain. The full-line min–max principle gives $\mu_j \geq \lambda_j$ because the λ_j are the first N eigenvalues, not an arbitrary selected cluster. Cyclicity of finite trace in (7) yields $\operatorname{tr} \widehat{Q} = \operatorname{tr} \Lambda + \operatorname{tr}(G^{-1}E)$. Each nonnegative eigenvalue displacement is at most their sum, proving (16) together with Corollary 4.1. Dirichlet min–max gives

$$\operatorname{Tr}(H_{V,I})_- \geq \sum_{j=1}^{N} (-\mu_j)_+ \geq -\sum_{j=1}^{N} \mu_j = \sum_{j=1}^{N} (-\lambda_j) - \tau_\chi.$$

The left side is also nonnegative. Finally, use (10), take traces after congruence by S, and diagonalize G. Each eigenvalue e_i of $I - G$ satisfies $e_i/(1 - e_i) \leq e_i/(1 - g)$. □

The trace alternative is particularly useful when a few states carry most of the lost mass. The norm estimate can be preferable when there are many comparable states: it uses the largest gradient eigenvalue rather than their sum. Both are available. The sharpness fixture in Example 3.2 is explicitly abstract; it cannot with $E = 0$ be a nontrivial localization of the first exact eigenvalue list, since its downward Ritz displacement would contradict (15). This is a useful physical realizability check on proposed finite matrix data.

Corollary 5.2 (All-exponent finite-list error)**.** *Put $b_j = -\lambda_j > 0$ and $\eta_* = \min\{\eta_\chi, \tau_\chi\}$. For any $\gamma > 0$,*

$$\operatorname{Tr}(H_{V,I})_-^\gamma \geq \sum_{j=1}^{N} (b_j - \eta_*)_+^\gamma. \tag{19}$$

The loss from the full-line first-N moment is at most

$$\mathcal{L}_\gamma = \sum_{j=1}^{N} [b_j^\gamma - (b_j - \eta_*)_+^\gamma] \leq \begin{cases} N\eta_*^\gamma, & 0 < \gamma \leq 1, \\ \gamma\eta_* \sum_j b_j^{\gamma-1}, & \gamma \geq 1. \end{cases} \tag{20}$$

At $\gamma = 1$, the aggregate loss τ_χ in (17) is preferable to the generic $N\eta_*$ bound when smaller.

Proof. Apply the decreasing function $x \mapsto (-x)_+^\gamma$ to the eigenvalue upper bounds and sum. If $0 < \gamma \leq 1$, subadditivity gives $b^\gamma - (b - \eta)_+^\gamma \leq \eta^\gamma$. If $\gamma \geq 1$, integrate the derivative $\gamma t^{\gamma-1}$ from $(b - \eta)_+$ to b; it is bounded by $\gamma\eta b^{\gamma-1}$. □

In the source range, these estimates combine with its imported Dirichlet upper bound to give, for $V = -W$,

$$\sum_{j=1}^{N} b_j^\gamma - \mathcal{L}_\gamma \leq L_\gamma \int_I W^p.$$

First fix N, then let the cutoff radius tend to infinity; only afterwards take the supremum of finite partial sums. The finite correction vanishes in this order. There is no uniform estimate for the omitted infinite list in (20). In particular, differentiating b^γ uniformly down to zero when $\gamma < 1$ would be invalid.

6 A weak-residual version for approximate states

The exact eigenfunction assumption is substantial. To make its replacement explicit, choose α so that

$$\|u\|_{\mathcal{V}}^2 = h_V[u,u] + \alpha \|u\|_2^2$$

is a positive complete form norm on $\mathcal{V} = \mathcal{D}_V$, continuously embedded in L^2. Let $J : \mathbb{C}^N \to \mathcal{V}$ have orthonormal L^2 columns ψ_j, and let $\Lambda = \operatorname{diag}(\lambda_j)$ be any real diagonal matrix. Define the form-dual residual map by

$$(\mathcal{R}c)(v) = h_V[v, Jc] - \langle v, J\Lambda c\rangle, \qquad v \in \mathcal{V}. \tag{21}$$

It is conjugate linear in v and linear in c. Let $J_{\chi^2} c = \chi^2 Jc$, and suppose certified bounds give

$$\|\mathcal{R}\|_{\mathbb{C}^N \to \mathcal{V}'} \le \rho, \qquad \left\|J_{\chi^2}\right\|_{\mathbb{C}^N \to \mathcal{V}} \le \beta.$$

No L^2 differential residual is presumed.

Proposition 6.1 (Defect-aware common-cutoff certificate)**.** *Form G, Q, E as in (4), now with the ψ_j, and assume $mI \le G \le MI$, $m > 0$. Then*

$$Q = \tfrac{1}{2}(\Lambda G + G\Lambda) + E + \tfrac{1}{2}(R + R^*), \qquad R_{ij} = (\mathcal{R}e_j)(\chi^2\psi_i). \tag{22}$$

The normalized form satisfies

$$\left\|G^{-1/2} Q G^{-1/2} - \Lambda\right\|_{\mathrm{op}} \le \left\|G^{-1/2} E G^{-1/2}\right\|_{\mathrm{op}} + \frac{\Delta(\sqrt{M} - \sqrt{m})^2}{4\sqrt{mM}} + \frac{\beta\rho}{m}. \tag{23}$$

Its trace differs from $\operatorname{tr}\Lambda + \operatorname{tr}(G^{-1}E)$ by at most $N\beta\rho/m$.

Proof. The product-rule proof of Lemma 2.1 still applies. The two weak pairings now have respective defects R_{ij} and $\overline{R_{ji}}$, proving (22). For unit coefficient vectors c, d,

$$|d^* R c| = |(\mathcal{R}c)(J_{\chi^2} d)| \le \beta\rho.$$

Hence $\|R\|_{\mathrm{op}} \le \beta\rho$, and the Hermitian part has at most the same norm. Congruence by $G^{-1/2}$ costs m^{-1}. Theorem 3.1 proves the norm estimate; the trace bound follows from $|\operatorname{tr} X| \le N \|X\|_{\mathrm{op}}$. □

The Ritz values therefore yield Dirichlet eigenvalue *upper* bounds relative to the ordered λ_j, with the error in (23). Approximate λ_j are not thereby certified lower bounds for the full spectrum. The nonnegative-displacement statement (15) cannot be reused unless the first-eigenvalue identification has independently been supplied. If only columnwise residual bounds ρ_j are available, $\rho \le (\sum_j \rho_j^2)^{1/2}$; the collective norm does not remove this cost by notation.

This is where the original Ram weak-operator analysis is needed: the dual norm is a supremum over the full form space, not just the chosen finite panel. Nonconforming or incorrectly traced states first require a conforming reconstruction or a trace lifting. Solving the compressed algebraic system does not certify ρ.

7 Exact examples and a global obstruction

Example 7.1 (Rational two-level normalization)**.** Take

$$G = \operatorname{diag}(9/16, 1), \quad \Lambda = \begin{pmatrix} -5 & 1 \\ 1 & -5 \end{pmatrix}, \quad E = 0.$$

Then

$$S = \operatorname{diag}(4/3, 1), \qquad \widehat{Q} = \begin{pmatrix} -5 & 25/24 \\ 25/24 & -5 \end{pmatrix}.$$

The error norm is $1/24$. The spectral width is 2, and (8) gives exactly $1/24$. The trace difference is zero, as required. The first-N physical realization caveat following Theorem 5.1 applies: the bottom Ritz level has moved downward. This verifies algebraic sharpness, not a Schrödinger construction.

Example 7.2 (An actual one-state Dirichlet enclosure)**.** At $\gamma = 1$, let

$$W(x) = 3\operatorname{sech}^2(2x), \qquad \phi(x) = \sqrt{2/\pi}\,\operatorname{sech}^{1/2}(2x).$$

Substitution $t = 2x$ gives $\|\phi\|_2 = 1$. Direct differentiation gives $H_{-W}\phi = -\phi$. Moreover $H_{-W} + 1 = (\partial_x + \tanh(2x))^*(\partial_x + \tanh(2x)) \geq 0$, so -1 is the ground energy. These are source profile facts, not new equality results.

Choose the angular cutoff with $w = R$. Since $\operatorname{sech}(2x) \leq 2e^{-2|x|}$,

$$g = 1 - \int \chi^2 |\phi|^2 \leq \frac{4}{\pi} e^{-2R}.$$

For R such that this bound is below one,

$$-1 \leq \lambda_1((-2R, 2R)) \leq -1 + \frac{\pi e^{-2R}}{R^2(1 - 4e^{-2R}/\pi)}. \tag{24}$$

Indeed $\Delta = 0$, and the exact Rayleigh quotient is $-1 + E/G$, with $E \leq \pi^2 g/(4R^2)$. At $R = 2$, the elementary bounds $e^4 > \sum_{j=0}^{7} 4^j/j! > 50$ and $3 < \pi < 22/7$ give the fully rational certificate

$$-1 \leq \lambda_1((-4, 4)) \leq -1 + \frac{121}{7154}. \tag{25}$$

Here $g \leq 2/75$ and $L^2 \leq 121/196$. This is a rigorous, deliberately nonoptimal enclosure obtained without a numerical eigenvalue solve. For a fixed source profile the explicit radius error is $O(R^{-2}e^{-2R})$; no such rate is asserted uniformly for arbitrary potentials.

Example 7.3 (Unseen wells cannot be certified away)**.** Fix the cutoff interval I, a length $a > 0$, and any integer M. Outside I, choose M pairwise disjoint intervals of length a, and put a potential $W = 2\pi^2/a^2$ on them and zero elsewhere. The normalized first Dirichlet sine on each interval, extended by zero, is an $H^1(\mathbb{R})$ trial vector. The vectors are orthogonal, have mutually zero cross forms, and each has Rayleigh quotient $-\pi^2/a^2$. Min–max therefore gives at least M negative eigenvalues, while every trial form supported in I sees zero potential. For each finite M, $W \in L^p$; its norm grows with M. Thus local data alone cannot bound an arbitrary omitted spectrum. A fixed global L^p mass constraint supplies additional information, but that is a separate input, not encoded in G, Q.

8 What the source's global argument still requires

The following dependency map distinguishes literal source reading from independent proof certification. All seven scientific modules were read; only the local results proved above are claimed as independently established contributions of this report.

Finite scalar action. The source's `action-framework.tex` associates to real orthonormal $u_i \in H_0^1(I)$ and $K = \operatorname{diag}(k_i) > 0$ the action

$$\mathcal{E}_K(a) = \int_I \{|a'|^2 + a^{\mathsf{T}} K^2 a - |a|^{2q}\}, \qquad a = Cu,$$

and claims the lower-triangular supremum is at least $\sum_i \int_{-k_i}^{k_i} (k_i^2 - s^2)^\sigma \, ds$. The triangular restriction matters: for ordered negative energies, the i-th row involves only the first i eigenfunctions, giving the required sign of $k_i^2 - k_j^2$. It is not an arbitrary matrix change of coordinates preserving every ordered spectral inequality.

Regularized matrix field. In `matrix-field.tex`, the parameter $\beta = \sigma - 1/2 \in (-1/2, 1/2)$ defines a regularized scalar concave function and a symmetric-cutoff integral field $M_\delta(B)$. The linear term odd in the integration variable cancels; the even remainder is integrable. Absolute convergence of the separate unsymmetrized leading term is not asserted. A trace primitive J_δ has gradient $-M_\delta$. Directional concavity gives scalar bounds, and coordinate identities apply when a coordinate vector is an eigenvector of both B and the fixed diagonal K. These statements are not arbitrary operator-concavity or simultaneous diagonalization principles.

The rank-one inequality. In `rank-one.tex`, the condition $M_\delta(B) = aa^{\mathsf{T}}$ forces positivity of the full symbol $K^2 - 2sB + s^2 I$, not positivity of $K^2 - B^2$ by itself. A resolvent tangent inequality and a single even probability measure give a trace-self-adjoint unital averaging map $Z \mapsto \int e^{itB} Z e^{-itB} \, d\nu(t)$. The rank-one identity $M^2 = |a|^2 M$ is then used to obtain

$$a^{\mathsf{T}}(K^2 - B^2)a \geq |a|^{2q}.$$

This is an essential source-specific nonlinear input. A finite Gram matrix or an arbitrary positive linear map does not supply it.

Connecting path and ordered limits. The module `connecting-path.tex` uses a cutoff ODE $\varepsilon B' = M_\delta(B) - aa^{\mathsf{T}}$, with endpoints $B(x_-) = K$, $B(x_+) = -K$. Its argument uses compactness of a joint zero set, row-sign symmetry, a free sign-group action, and parity of a one-dimensional deformation after smooth approximation. Every row of the shooting matrix must be nonzero; full invertibility of that matrix is not established by this statement or needed for the sign action. Directional barriers remove the auxiliary cutoff. None of these global shooting or transversality steps follows from a small finite residual in the displayed ODE.

In `action-limit.tex`, compactness is at fixed $\delta > 0$. The term $\varepsilon^{-1} \| M_\delta - aa^{\mathsf{T}} \|_{\mathrm{HS}}^2$ forces an algebraic constraint in a maximizing contradiction sequence. One first lets $\varepsilon \downarrow 0$, then $\delta \downarrow 0$; no quantitative, jointly uniform shooting rate is supplied. Our cutoff constants do not provide one.

Spectral passage. The final module `spectral.tex` proves form bounds and compactness, applies the action estimate on bounded intervals, and takes $R \to \infty$ at fixed N, followed by the supremum over N. Its generic estimate (6) is already valid. Theorems 3.1 and 5.1 refine only this last finite-family passage. The source's profile calculation gives attainment of its constant, rather than a classification of all optimizers. No correction to these retained mechanisms is asserted here; their global use remains an imported claim, not certified by our finite tests.

9 Original Ram connections and precise scope

Three original private proof passages were freshly read locally. The physical-metric chapter [5] constructs a measured Gram form and insists that compatible changes act by congruence on both mass and energy. That is exactly why SQS, rather than a coordinate matrix with an unverified Euclidean norm, occurs here. Its distinction between a full physical residual and its measured projection is used in Proposition 6.1. The new increment is not the existence of a Gram normalization: it is the exact common-cutoff identity, its spectral-width double commutator, and the first-list trace argument that exploit this source's particular eigenfunctions.

The spectral-limits chapter [6] separates strong from norm convergence, includes a spectral-pollution example for an indefinite operator, and proves the role of form-dense Rayleigh spaces in semibounded problems. We do not transfer its indefinite example into a false claim that every semibounded conforming approximation pollutes. Here the precise conforming min–max inequality is what justifies the one-sided enclosures; it does not enumerate the unseen full-line spectrum.

The original weak-boundary treatment [7] defines operators through weak pairings, distinguishes Dirichlet and Neumann data, and uses a full dual residual for coercive error control. Its later Galerkin discussion preserves both form and mass under congruence and separates quadrature error from exact variational identities. These are useful domain checks for rough potentials and for the residual extension here. No Neumann compatibility or gauge is silently imposed on this Dirichlet exhaustion problem.

The neighboring matrix-equality analysis is not being credited again: Young defects, Cayley evolution, and moving range projectors are not the new results of this scalar report. Nor are classical IMS localization, min–max, or commutator norm inequalities claimed as new Ram principles. The contribution is their fully quantified combination at a specific retained step, with the sharp abstract coefficient, trace cancellation, and actual cutoff example proved above.

10 Literature, evidence, and limitations

As checked on October 9, 2026, Read and Schulz's primary arXiv version record and abstract [2] state the sharp one-dimensional $\gamma = 1$ constant $4/(3\sqrt{3}\pi)$, together with an operator-valued extension. That abstract is not a proof of every $1/2 < \gamma < 3/2$, and its full proof was not independently audited for this report. The source [1] claims the wider scalar interval. The publisher's selected-formalization description [4] also describes the scalar full negative moment and attainment. Reading that description is not execution of Lean or verification of its full dependency graph.

The fresh public proof-level comparison [3, Appendix A] supplies the classical quadratic-form IMS identity for Lipschitz multipliers. Only its relevant appendix passage and opening metadata were inspected, not the full helium-system proof. No historical priority is claimed for the local corollaries in this report. A failed attempt to open an additional textbook PDF is recorded as failed access, not as proof reading.

The literal local source scope is all 11 cached text files, 1,672 lines including 1,332 lines in the seven scientific modules. The fresh original-private scope is 589 lines: the full 210-line physical-metric chapter, the full 242-line spectral-limits chapter, and lines 81–160 and 492–548 of the 1,066-line original weak-boundary manuscript. Its complementary 929 lines and its external figures were not inspected. Bibliographic inclusion is not a claim to have read every cited work, and no whole-corpus audit is asserted. The accompanying read ledger records pre-read hashes, literal display intervals, unread complements, and post-read hashes. Captured-tree byte identity is provenance only.

The executable local checks concern matrix identities, exact positive-semidefinite bounds, polynomial weak pairings, moment losses, and the explicit rational enclosure. They are supplemental fixtures, not validation of a continuum spectral theorem, global action path, or an infinite negative-eigenvalue sum. Their receipt distinguishes parameter cases from assertions; independent external checks are not included in the owned counts.

11 Benefits, costs, and further work

Benefits. The common-cutoff Gram cost is shift invariant and quadratic, not linear, in small lost mass. Equal-energy blocks have no commutator cost. Angular cutoffs control gradient leakage with the same lost mass, while the exponent-one trace bound accounts for an entire fixed list without a spectral-width penalty. One may take the smaller of trace and operator bounds; neither is uniformly better. The weak residual variant states exactly which additional full-domain certificate an approximate input requires.

Costs and failure modes. True eigenfunctions, certified mass tails, and physical form integrals are substantive inputs. The condition $G > 0$ and its smallest eigenvalue are unavoidable conditioning costs. The bounds can deteriorate with N, the spectral width, or a weakly confined last state; no dimension-uniform rate follows from fixed-family exhaustion. Near zero, the moment map for $\gamma < 1$ is only Hölder continuous, as reflected in the loss bound. Finite residual sampling is not a dual-norm bound. Example 7.3 rules out unconditional spectral completion from a bounded observation window.

Feasible extensions. A useful next step is an independently validated tail enclosure for a specified potential, combined with conforming interval eigenfunctions and a full weak-residual estimate. Clusterwise cutoffs could reduce spectral width, but cross-cluster mass and form terms would have to be retained rather than discarded. Direct certification of $\operatorname{tr}(G^{-1}E)$ may be preferable to an operator norm when only a first moment is required. Optimizing cutoff width against a known tail profile is a one-variable problem after the profile has been justified; it is not an unmeasured computational speedup. Extending these certificates to a complete spectral remainder would require additional global information absent from the present hypotheses.

Conclusion. The source's physical scalar setting allows more structure than a generic Gram perturbation. Retaining exact localization yields a sharp abstract quadratic spectral-width bound and a separate trace-exact finite-list estimate, with fully stated domain and residual gates. These are notable local quantitative improvements, independent of applicability level L3. They neither prove a new sharp global Lieb–Thirring constant nor independently certify the source's connecting-path construction.

References

[1] OpenAI, *Sharp one-dimensional Lieb–Thirring constants*, September 23, 2026. Cached manuscript and seven scientific modules in the public `openai/math` repository; all local text read, global result treated as an imported source claim.

[2] L. Read and M. R. Schulz, *The sharp one-dimensional Lieb–Thirring inequality for the sum of eigenvalues*, arXiv:2609.10478v2, September 14, 2026. Abstract and version record checked; full proof not independently audited here.

[3] D. Hundertmark, M. Jex, and M. Lange, *Quantum Systems at The Brink: Helium–Type Systems*, Archive for Rational Mechanics and Analysis **250**, 67 (2026), doi:10.1007/s00205-026-02167-7. Relevant proof read: Appendix A, especially Lemma A.3.

[4] OpenAI, selected formalization description, `lean/docs/262.md`. Publisher scope description read; no local Lean execution or independent formal dependency audit performed.

[5] M. Subbarao, *RamCalBooks*, volume 21, original chapter `ch01d-physical-metrics.tex`. Confidential local manuscript, full 210-line chapter freshly read.

[6] M. Subbarao, *RamCalBooks*, volume 21, original chapter `ch01f-spectral-limits.tex`. Confidential local manuscript, full 242-line chapter freshly read.

[7] M. Subbarao, original `TR8-lin-non-lin.tex`, *RamNonlin-IE-PDE-astra* corpus. Confidential local manuscript; fresh proof scope limited to lines 81–160 and 492–548.